

SMART INDIA HACKATHON 2026



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HACKATHON
2026

RESCUE**MESH**

Problem Statement ID – SIH26206

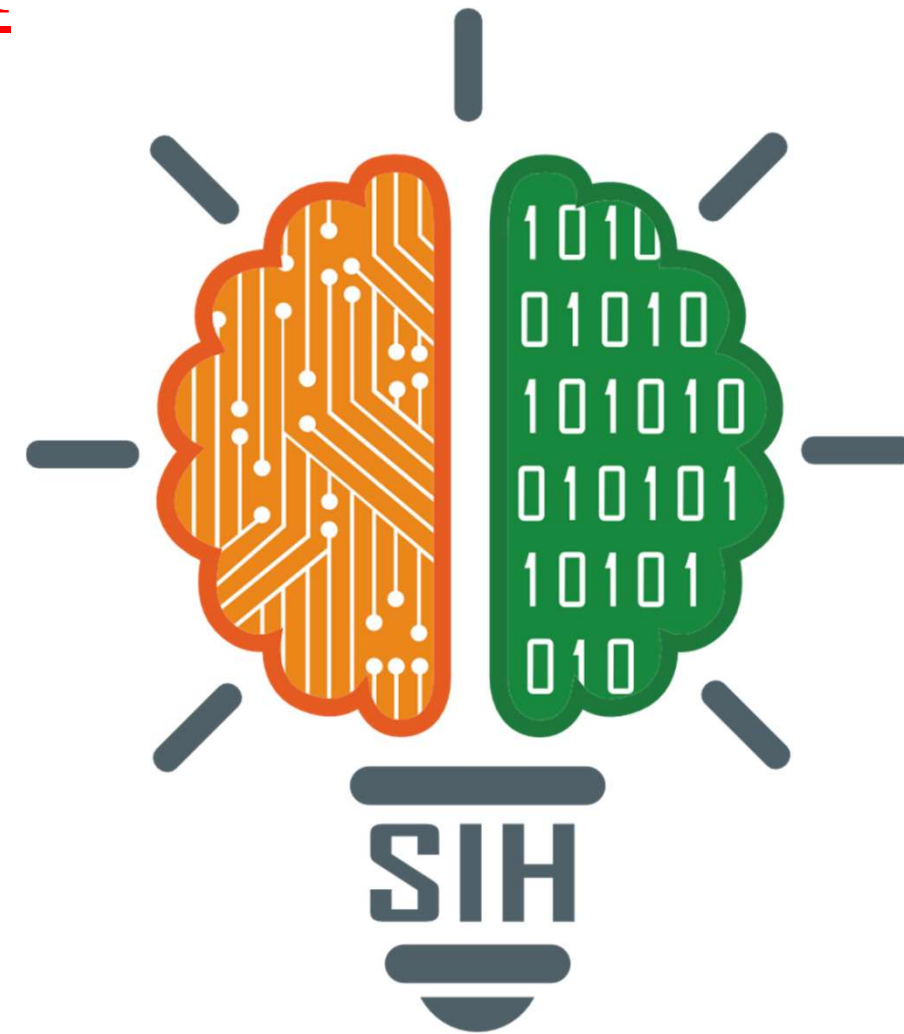
Problem Statement Title- Student Innovation-Disaster management includes ideas related to risk mitigation, planning and management before,after or during a disaster.

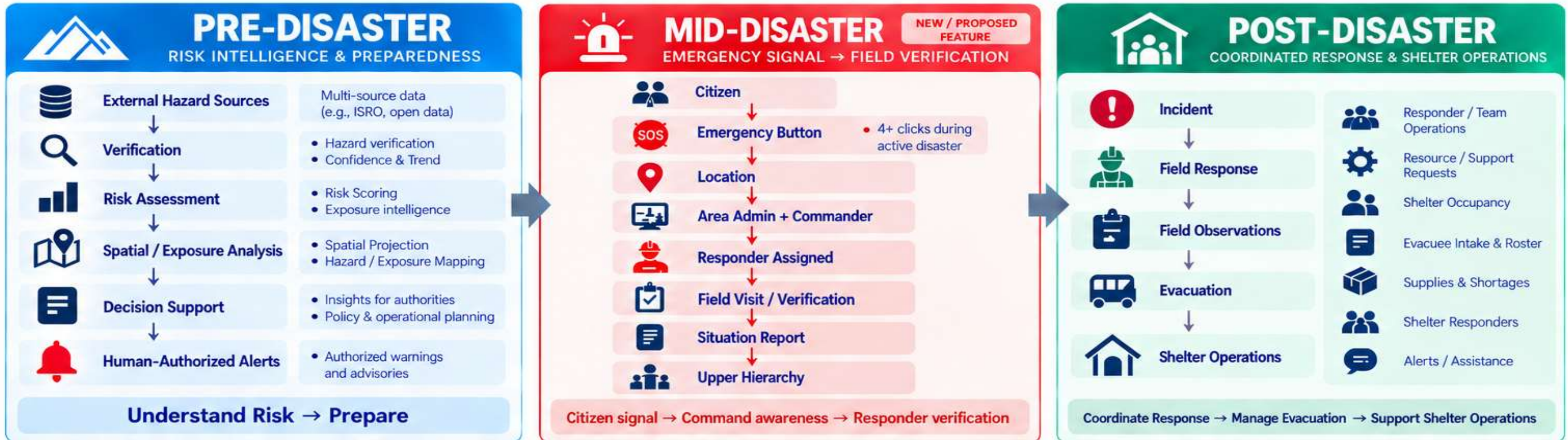
Theme- Disaster Management.

PS Category- Software.

Team ID- 143620

Team Name- Didex.





HOW RESCUEMESH ADDRESSES KEY PROBLEMS



1. DATA ISOLATION

Problem
Fragmented information & verbal relays

RescueMesh Response
Shared incident / intelligence context



2. COORDINATION GAPS

Problem
Command and field coordination can become fragmented

RescueMesh Response
Connected operational workflow



3. SHELTER OPERATIONS

Problem
Occupancy, evacuees, supplies and responders become difficult to coordinate

RescueMesh Response
Integrated shelter operations



4. RESPONSE DELAY

Problem
Delayed awareness and field verification

RescueMesh Response
Structured emergency reporting + responder workflow

WHAT MAKES RESCUEMESH DIFFERENT?



1. PRE → MID → POST

One connected disaster lifecycle instead of a response-only workflow.



2. PEOPLE-CENTRIC APPROACH

Bridges citizens, authorities and field responders on one platform.



3. REAL-WORLD DATA INTEGRATION

Uses trusted, open and official sources with verification and spatial intelligence.

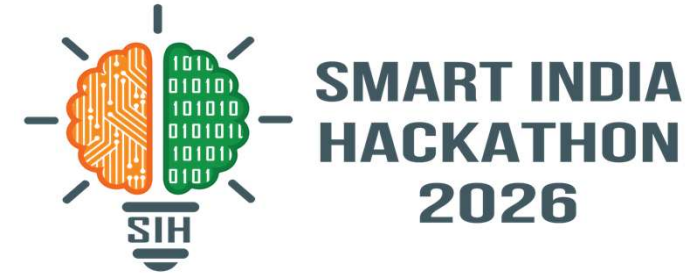


4. HUMAN CONTROL & AUDITABILITY

Human-authorized alerts, role-based access and traceable actions.

DIDEX

TECHNICAL APPROACH



TECHNOLOGIES USED

Open, reliable and scalable technologies for a secure and impactful solution.



Frontend (Web Application)



React 18



Vite 6



Tailwind CSS



Leaflet / MapLibre



Backend (API & Services)



Python 3.14



FastAPI



Pydantic v2



SQLAlchemy 2



Database



SQLite (local)



PostgreSQL (production)



Security & Identity



JWT



RBAC



Argon2id



TOTP 2FA



Audit Logs



Geospatial & Mapping



GeoJSON / spatial analysis



Authoritative Data Sources



CWC



USGS



NDMA

Real-time & official data for accurate risk intelligence



IMPLEMENTATION METHODOLOGY

Deterministic, modular and auditable risk intelligence pipeline (Steps 1-10)



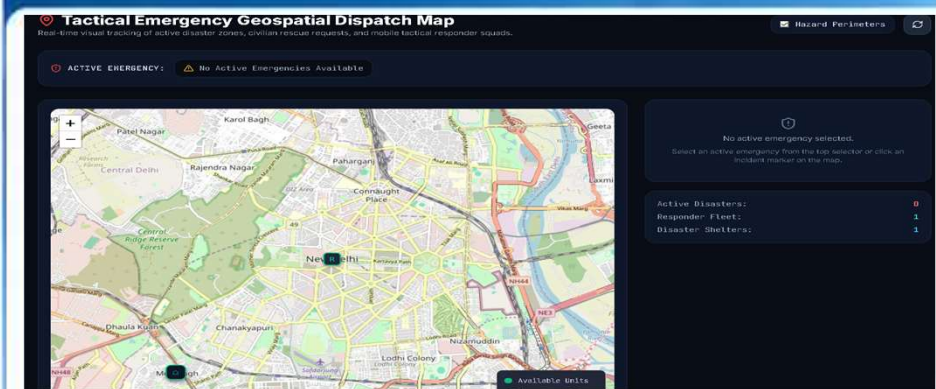
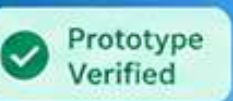
OPERATIONAL FLOW: FROM DISTRESS TO RESPONSE

Citizen reports are processed, verified and dispatched through a controlled workflow.

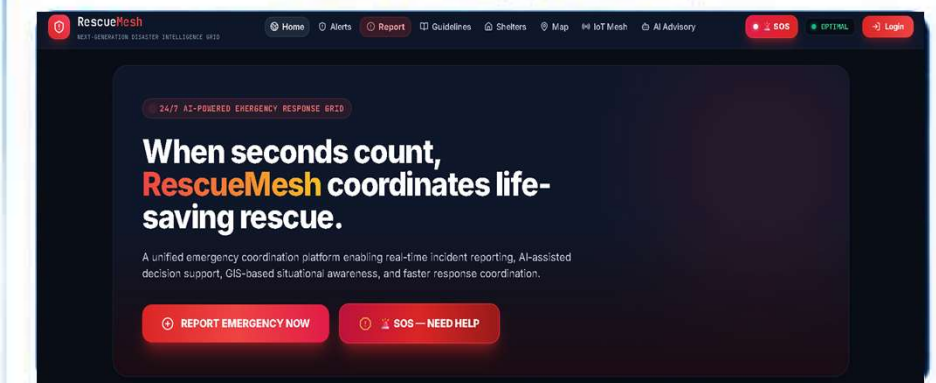


WORKING PROTOTYPE

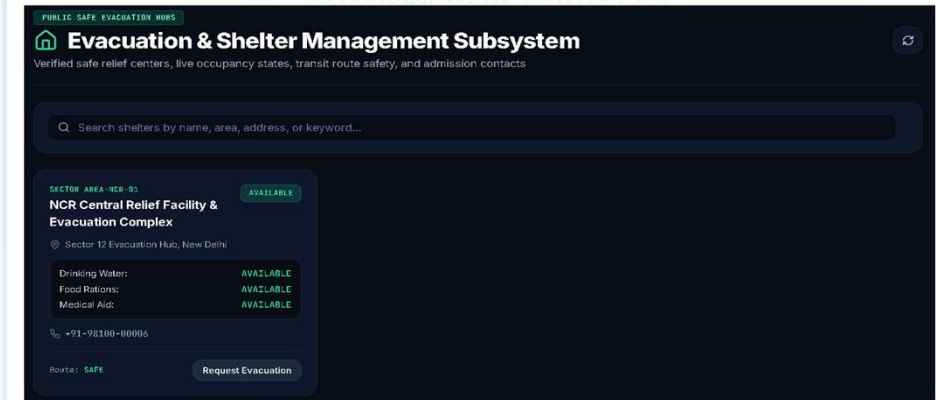
Live workflow screens from RescueMesh



1. Public Risk Map (Leaflet GIS)



2. Emergency Report Form



3. Evacuation & Shelter Dashboard



CORE ENGINEERING PRINCIPLES

Guiding every component of RescueMesh



Deterministic

Rule-based, consistent and reliable



Explainable

Transparent logic and traceable decisions



Role-scoped

Access and actions as per role (RBAC)



Geo-aware

Built on spatial data and location intelligence



Auditable

Persistent logs for every action



Scalable

Modular design for future expansion



Real-world Impact

Built for safer people and resilient communities

FEASIBILITY & VIABILITY

FEASIBILITY ANALYSIS

% TECHNICAL & OPERATIONAL FEASIBILITY

- **Tech Stack:** AI-assisted analysis, Leaflet GIS, FastAPI + PostgreSQL/PostGIS, workflows, web + database platform. GIS coordination.
- **Operational:** Automated Incident, Evacuation & Shelter workflows. Incident management.
- **Public Interface:** Disaster Guidelines access & multi-lingual reporting/nearest shelter locator. Emergency reporting.

📈 ECONOMIC & SUSTAINED VIABILITY

- **Financial/Cost:** 100% Open-Source stack = Zero software licensing costs.
- **Scalability:** Cloud-native microservices for high multi-district disaster load.
- **Sustainability:** NDMA IRS compliant for fast government & ground adoption.

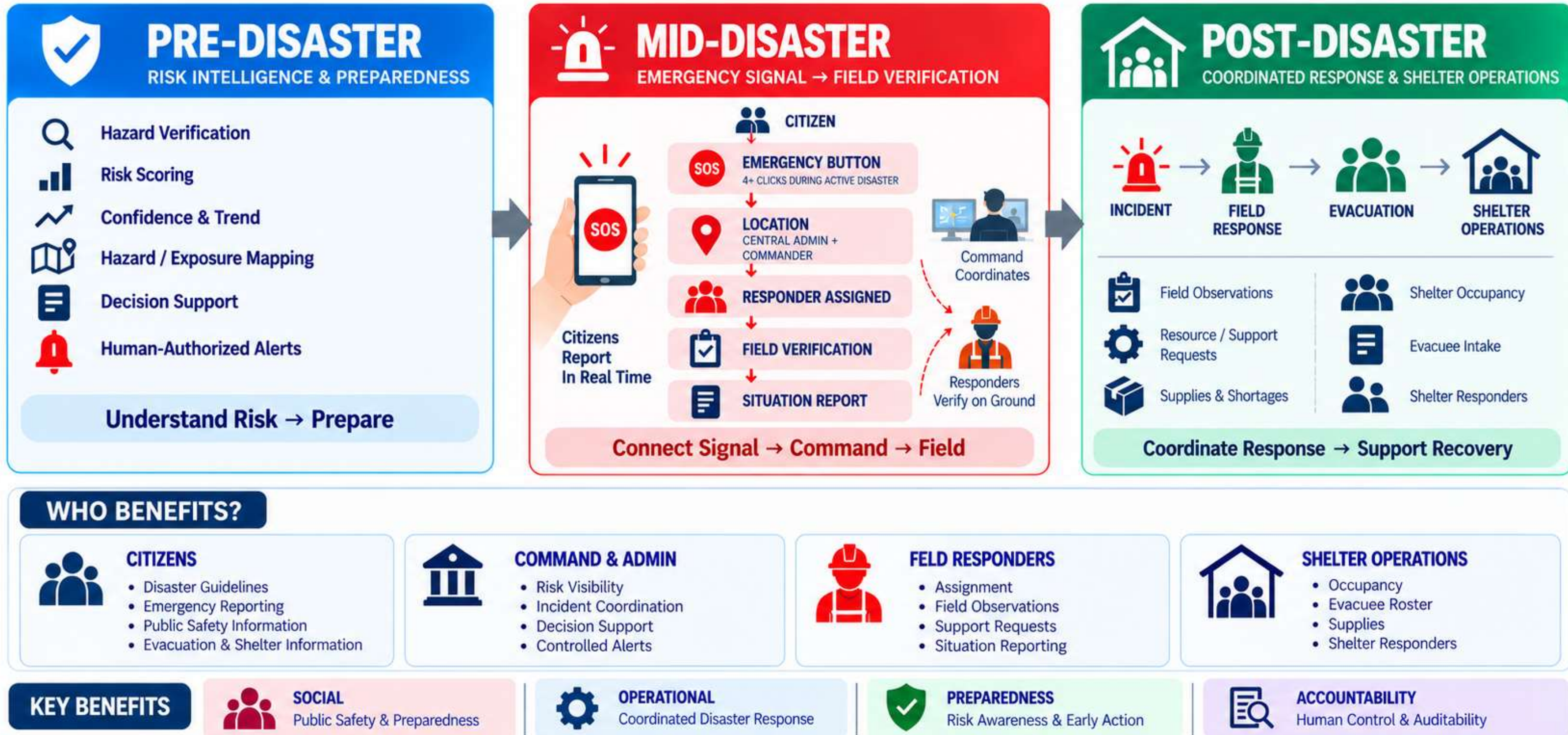
🔵 **High feasibility + Low-cost long-term viability for NDMA/SDMA.**

⚠️ CHALLENGES & RISKS

- 🗄️ **Data Accuracy**
Incomplete or incorrect emergency reports during high chaos.
- 📍 **Map & Service Dependency**
Dependence on external GIS visualization map servers.
- 🔗 **Integration & Interoperability**
Different data sources and APIs may use incompatible formats or standards.
- 🛡️ **Access & Security Risk**
Unauthorized access risking critical operational controls.
- 🚨 **Emergency Data Load**
Simultaneous disaster reports causing server bottlenecks.
- 📉 **Fake Report Verification**
Unverified public distress signals overwhelming rescue teams.

🛡️ MITIGATION STRATEGIES

- ✅ **Input Validation + AI filtering**
Input validation + AI filtering for duplicate report merging.
- ✅ **Configurable Map & Service Setup**
dependence on external GIS visualization map servers, dependence on external GIS visualization map servers, map service setup. Flexible fallback to configurable map tiles + offline tile caching.
- ✅ **RBAC & Audit Logging**
Role-Based Access Control, JWT AUTH : an immutable audit logging.
- ✅ **Modular APIs & Microservices**
dependence on external GIS visualization map servers. Async FastAPI endpoints + Redis cache, with modular APIs, database workflows, and testing.
- ✅ **Human/area verification**
Area-level ground-team approval workflow before operational processing.





POLICY REFERENCE

Guidelines, Standards & Frameworks



The Sphere Handbook

(Humanitarian Charter & Minimum Standards in Disaster Response)

<https://spherestandards.org/handbook-2018/>



NDMA – National Disaster Management Guidelines

<https://ndma.gov.in/guidelines>



YUVA APDA Training Scheme (NDMA)

<https://aapdamitra.ndma.gov.in/home>



Sendai Framework for Disaster Risk Reduction (UNDRR)

<https://www.undrr.org/sendai-framework>



Aligned with national & global standards for disaster management.



ENGINEERING REFERENCE

Tools, APIs & Technical Resources



Geospatial Data Interoperability (GEOJSON Standard)

<https://geojson.io/>

Spatial data formatting for map visualization and integration.



Multi-Factor Authentication (TOTP Algorithm)

<https://totp.app/>

Secure user authentication for citizen and admin access.



PostgreSQL + PostGIS

<https://www.postgresql.org/>

Robust open-source database with spatial extension for GIS data.



FastAPI (Backend Framework)

<https://fastapi.tiangolo.com/>

High-performance APIs for real-time services.



Open-source, scalable and secure technology stack.



EMPIRICAL BENCHMARK & HUMANITARIAN DATA

Real-World Data & Case Insights



Disaster Event Databases

- EM-DAT (CRED): <https://www.emdat.be/>
- ReliefWeb: <https://reliefweb.int/>

Historical disaster data for analysis and model validation.



Response Cohesion

Recommended measures to build national and community response capacities to warning messages.



Logistic Optimization

Optimization of transportation, warehousing and distribution to lower costs and improve efficiency.



NGO & Field Reports

Case studies and field reports from real disaster responses for practical insights.



Evidence-driven insights from real events and humanitarian domains.



ADDITIONAL REFERENCES

Datasets, APIs & Useful Links



Weather & Hazard Data (IMD)

<https://mausam.imd.gov.in/>

Cyclone, rainfall, and real-time weather warnings.



Bhuvan (ISRO GIS Platform)

<https://bhuvan.nrsc.gov.in/>

Satellite imagery and geospatial layers for risk mapping.



OpenStreetMap

<https://www.openstreetmap.org/>

Open-source base maps for visualization.



Government Open Data Portal

<https://data.gov.in/>

Access to multiple official datasets.



Reliable and open sources for real-time and historical data.

PROTOTYPE LINK-<https://rescuemesh-upgr.onrender.com>

GIT HUB LINK- <https://github.com/sankalprai-IoT/RescueMesh-SIH-2026>